

Cluster Theory

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Abstract

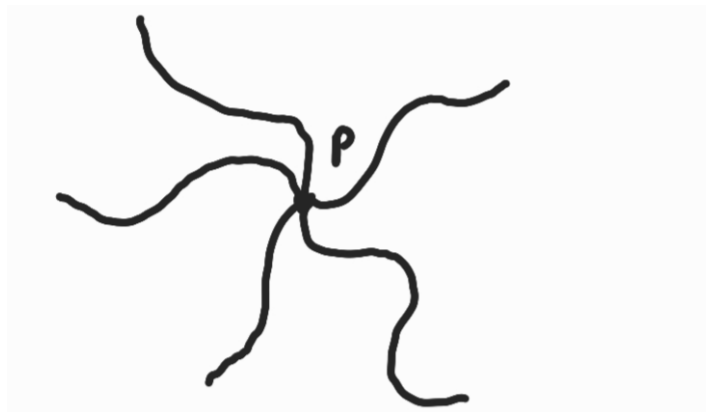
This preliminary note introduces an informal way to describe groups of irregular lines that are joined together. A *cluster* is a group of lines connected through other lines, and a *tie* records a joining within a component. The examples and notation are intended to communicate the basic idea visually and can be developed further using formal graph-theoretic definitions.

1 Basics of Cluster Theory

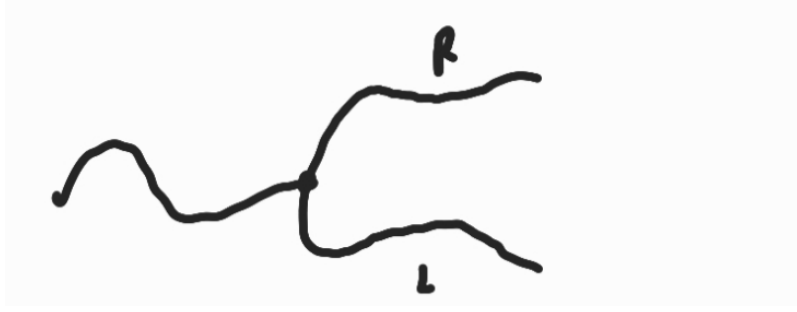
Recall that Intangible Shapes connects points irregularly.



The main idea of Cluster Theory is that one may have a single point P with a finite number of irregular lines, where the irregular lines are finite as well.

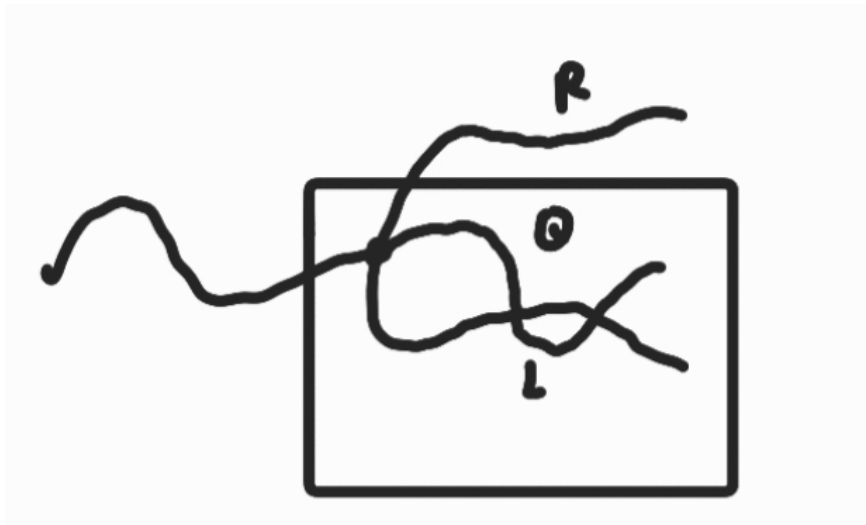


Consider the following example.

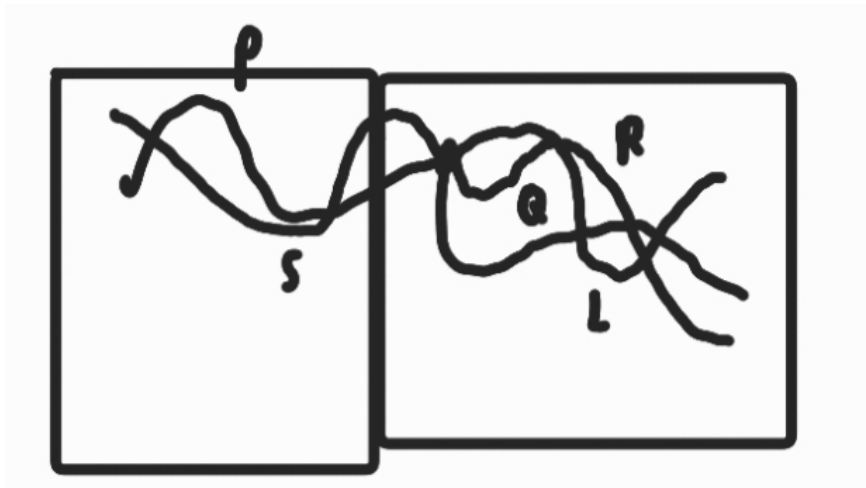


Let another irregular line Q be tied to L . As shown below, L and Q , the boxed lines, may be treated as a single component. If R is also tied to L and Q , they become a single component:

$$P \text{ (Group) Component}_{RQL} = \text{Clusters.}$$



What if another irregular line S is tied to P ?



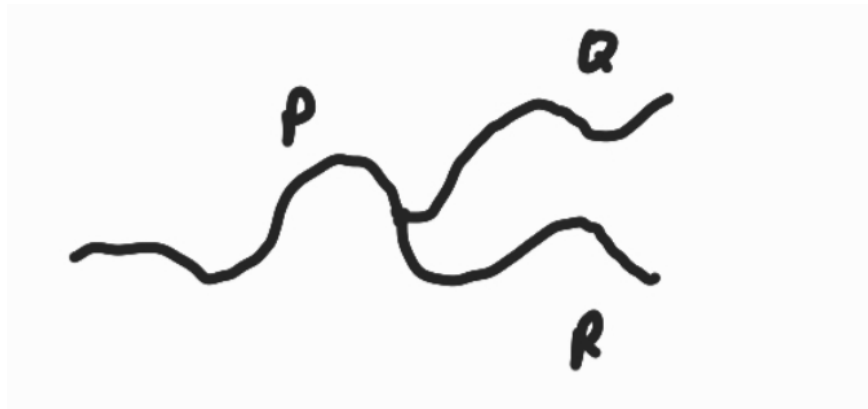
The result can be written as

$$\text{Component}_{PS} \text{ (Group) Component}_{RQL} = \text{Clusters.}$$

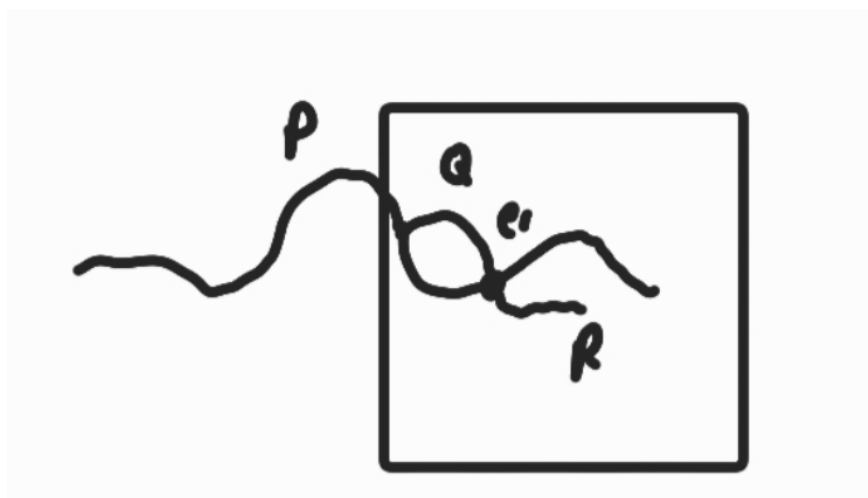
Definition 1. Given irregular lines P , Q , R , L , and S , a cluster is a group of lines that touch each other through other irregular lines.

2 Tying in a Component

Consider the following example.



Let us tie Q and P and treat them as a single component. Label the first tie as e_1 .



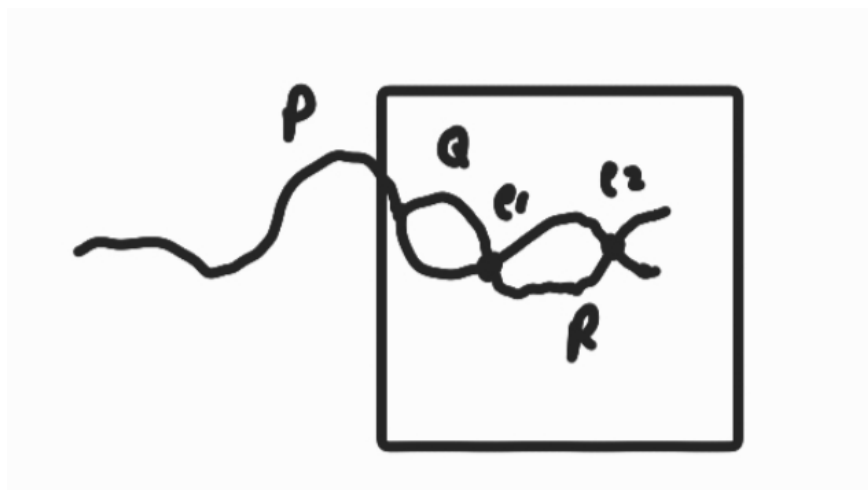
The equation is

$$P \text{ (Group) Component}_{QR} = \text{Cluster.}$$

Where will we include the tie in the equation?

$$P \text{ (Group) Component}_{QR} [e_1] = \text{Cluster.}$$

Now, what if another tie e_2 is added?



The equation then becomes

$$P \text{ (Group) Component}_{QR} [e_1, e_2] = \text{Cluster.}$$

3 Explaining the Notation

Consider the expression

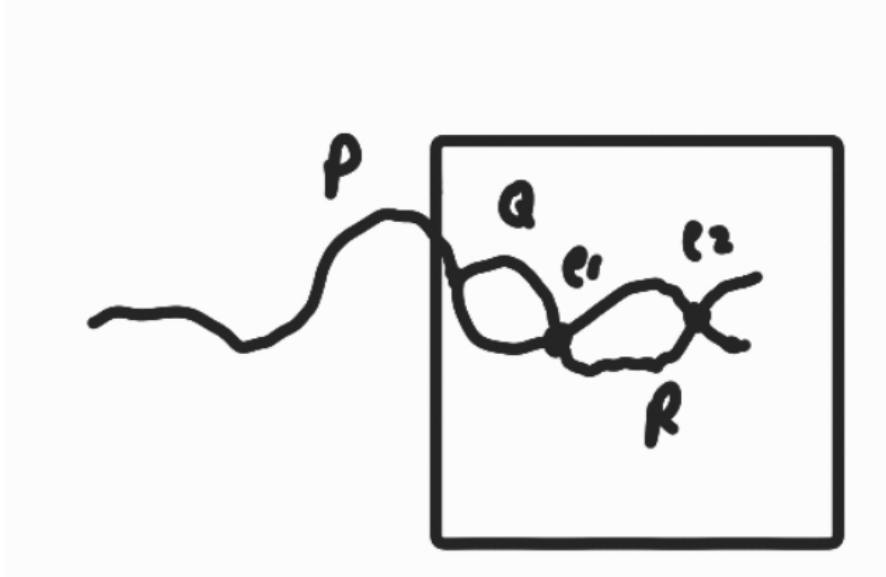
$$P \text{ (Group) Component}_{QR} [e_1, e_2] = \text{Cluster.}$$

The left-hand side is explained as follows:

- P stands for a single irregular line.
- (Group) means that the irregular line and the components are grouped because they share the same point.
- Component_{QR} indicates that irregular lines Q and R are tied and are therefore treated as a single component.
- $[e_1, e_2]$ indicates the number of ties in a single component.

4 Connecting Two Clusters

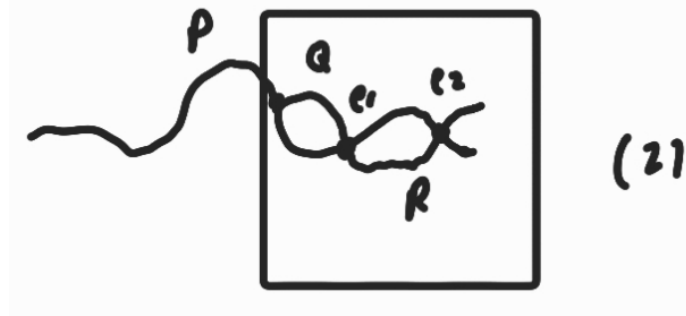
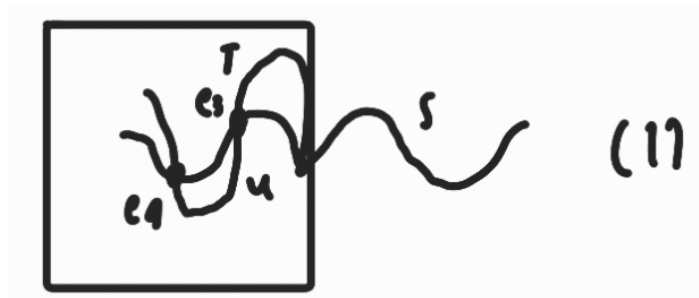
Consider the following cluster:



Its equation is

$$P \text{ (Group) Component}_{QR} [e_1, e_2] = \text{Cluster.}$$

Now introduce another cluster.

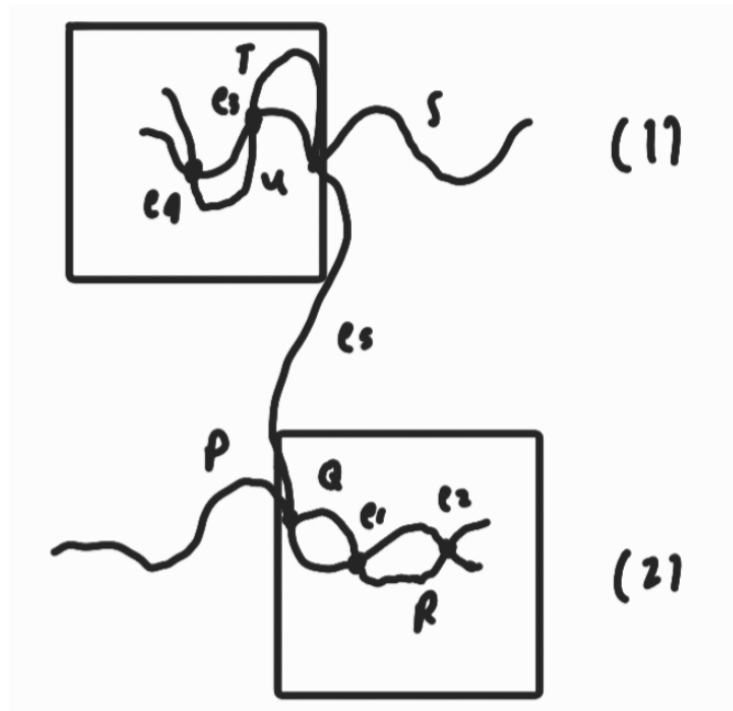


The two clusters are written separately as

$$C_1 = S \text{ (Group) Component}_{TU} [e_3, e_4],$$

$$C_2 = P \text{ (Group) Component}_{QR} [e_1, e_2].$$

Suppose a new irregular line, labelled e_5 , ties C_1 to C_2 .



The two clusters are not equal to each other. Instead, e_5 joins them to form one larger cluster:

$$\text{Cluster} = (S \text{ (Group) Component}_{TU} [e_3, e_4])$$

$$\cup (P \text{ (Group) Component}_{QR} [e_1, e_2]) \cup \{e_5\}.$$

5 Graph-Theoretic Definitions

The ideas in Cluster Theory can be described formally using a finite undirected multigraph

$$G = (V, E).$$

Each vertex in V represents an irregular line, and each edge in E represents a tie between two irregular lines. Parallel edges are allowed, so more than one tie between the same pair of irregular lines can be recorded separately.

Definition 2. *A cluster is a connected component of G . Equivalently, two irregular lines belong to the same cluster if and only if there is a path of ties between their corresponding vertices.*

For the two clusters introduced above, let

$$\begin{aligned} C_1 &= (V_1, E_1), & V_1 &= \{S, T, U\}, & E_1 &= \{e_3, e_4\}, \\ C_2 &= (V_2, E_2), & V_2 &= \{P, Q, R\}, & E_2 &= \{e_1, e_2\}. \end{aligned}$$

Because C_1 and C_2 are distinct clusters, their vertex sets are disjoint:

$$V_1 \cap V_2 = \emptyset.$$

Let $e_5 = \{x, y\}$ be a new tie, where $x \in V_1$ and $y \in V_2$. After this tie is added, the resulting graph is

$$G' = (V, E \cup \{e_5\}).$$

The tie e_5 creates a path between the two former clusters. Therefore, they become one larger connected component:

$$C = (V_1 \cup V_2, E_1 \cup E_2 \cup \{e_5\}).$$

Thus,

$$C = (\{S, T, U, P, Q, R\}, \{e_1, e_2, e_3, e_4, e_5\}),$$

or, equivalently,

$$C = C_1 \cup C_2 \cup \{e_5\}.$$

In this graph-theoretic notation, e_5 denotes the tie, or edge, that joins the two clusters. If e_5 is intended to represent a new irregular line rather than a tie, it should instead be represented by a new vertex with two incident tie edges.

6 Connectivity Modeling for Engineering Systems

Cluster Theory may be applied to engineering systems by representing a system as a finite undirected graph:

$$G = (V, E).$$

Each vertex represents an engineering element, such as a wire, circuit module, sensor, pipe segment, machine, or communication device:

$$V = \{v_1, v_2, \dots, v_n\}.$$

Each edge represents a physical or logical connection between two elements:

$$E \subseteq \{\{v_i, v_j\} : v_i, v_j \in V, i \neq j\}.$$

A cluster is a connected component of G :

$$C_k = (V_k, E_k).$$

Two elements u and v are connected when there exists a path of ties between them:

$$u \rightsquigarrow v.$$

The complete engineering system is connected when it has only one connected component:

$$|\text{ConnComp}(G)| = 1.$$

If two separate subsystems

$$C_1 = (V_1, E_1) \quad \text{and} \quad C_2 = (V_2, E_2)$$

are joined by a new tie

$$e_5 = \{x, y\}, \quad x \in V_1, \quad y \in V_2,$$

then the combined system is

$$C = (V_1 \cup V_2, E_1 \cup E_2 \cup \{e_5\}).$$

A tie is critical if its removal separates a cluster into two or more components:

$$|\text{ConnComp}(G - e)| > |\text{ConnComp}(G)|.$$

Such a tie is called a *bridge*. In engineering, a bridge can represent a single connector, wire, pipe, or communication link whose failure disconnects part of the system.

Additional engineering properties may be assigned to each tie:

$$e_i = (u_i, v_i, \ell_i, r_i, p_i),$$

where

$$\ell_i = \text{length}, \quad r_i = \text{resistance or cost}, \quad p_i = \text{failure probability}.$$

For a path

$$P = (e_1, e_2, \dots, e_m),$$

its total resistance or cost is

$$R(P) = \sum_{e_i \in P} r_i.$$

The lowest-cost connection between two elements u and v is

$$P^* = \arg \min_{P: u \rightsquigarrow v} \sum_{e_i \in P} r_i.$$

Therefore, Cluster Theory can model how engineering elements form connected systems, how separate subsystems are joined, and which ties are essential for system reliability.

References

- [1] Erwin Kreyszig. *Advanced Engineering Mathematics*. 10th ed., John Wiley & Sons, 2011.
- [2] Richard Hammack. *Book of Proof*. 2nd ed., Richard Hammack, 2013.